Linear Equations in Two Variables

The big idea

An equation like **2x + 3y = 12** has two unknowns. Unlike x + 3 = 5, it doesn't have a single answer: it has **infinitely many** solutions, and when you plot them they form a **straight line**.

STANDARD FORM

ax + by + c = 0, where a, b, c are real numbers and a and b are not both zero.

WRITE IN STANDARD FORM AND FIND A, B, C: 2X = −5Y

2x + 5y + 0 = 0 ⇒ a = 2, b = 5, c = 0.

Finding solutions

Choose any value for x, then solve for y (or the other way round).

FIND FOUR SOLUTIONS OF X + 2Y = 6

x = 0 ⇒ y = 3; x = 2 ⇒ y = 2; x = 4 ⇒ y = 1; x = 6 ⇒ y = 0.

Solutions: (0, 3), (2, 2), (4, 1), (6, 0) … and infinitely more.

CHECK WHETHER (2, 1) IS A SOLUTION OF 3X − 2Y = 4

3(2) − 2(1) = 4 ✓. Yes, it is a solution.

Graph of a linear equation

Every solution is a point on one straight line, and every point on that line is a solution. So to draw the graph, plot **two or three** solutions and join them.

- $y = 3$ (or $0 \cdot x + y = 3$) is a line **parallel to the x-axis**.
- $x = -2$ is a line **parallel to the y-axis**.
- $y = mx$ passes through the **origin**.

WORD PROBLEMS

"The cost of a notebook is twice the cost of a pen": let pen = ₹x, notebook = ₹y, then **$y = 2x$**, or $2x - y = 0$.

Quick recap

- $ax + by + c = 0$ has infinitely many solutions.
- Each solution is a point (x, y); together they form a straight line.
- Plot two or three points to draw the graph.

Practice questions

1. How many solutions does $2x + 3y = 6$ have?
   (a) One (b) Two (c) Infinitely many (d) None
2. Which point is a solution of $x + y = 5$?
   (a) (2, 3) (b) (2, 2) (c) (3, 3) (d) (1, 5)
3. The graph of $y = 4$ is a line
   (a) parallel to the x-axis (b) parallel to the y-axis (c) through the origin (d) along the y-axis
4. In $3x - 5y + 7 = 0$, the value of b is
   (a) 3 (b) −5 (c) 7 (d) 5
5. If (k, 2) is a solution of $2x + y = 8$, then k is
   (a) 2 (b) 3 (c) 4 (d) 6

Answers: 1. c 2. a 3. a 4. b 5. b